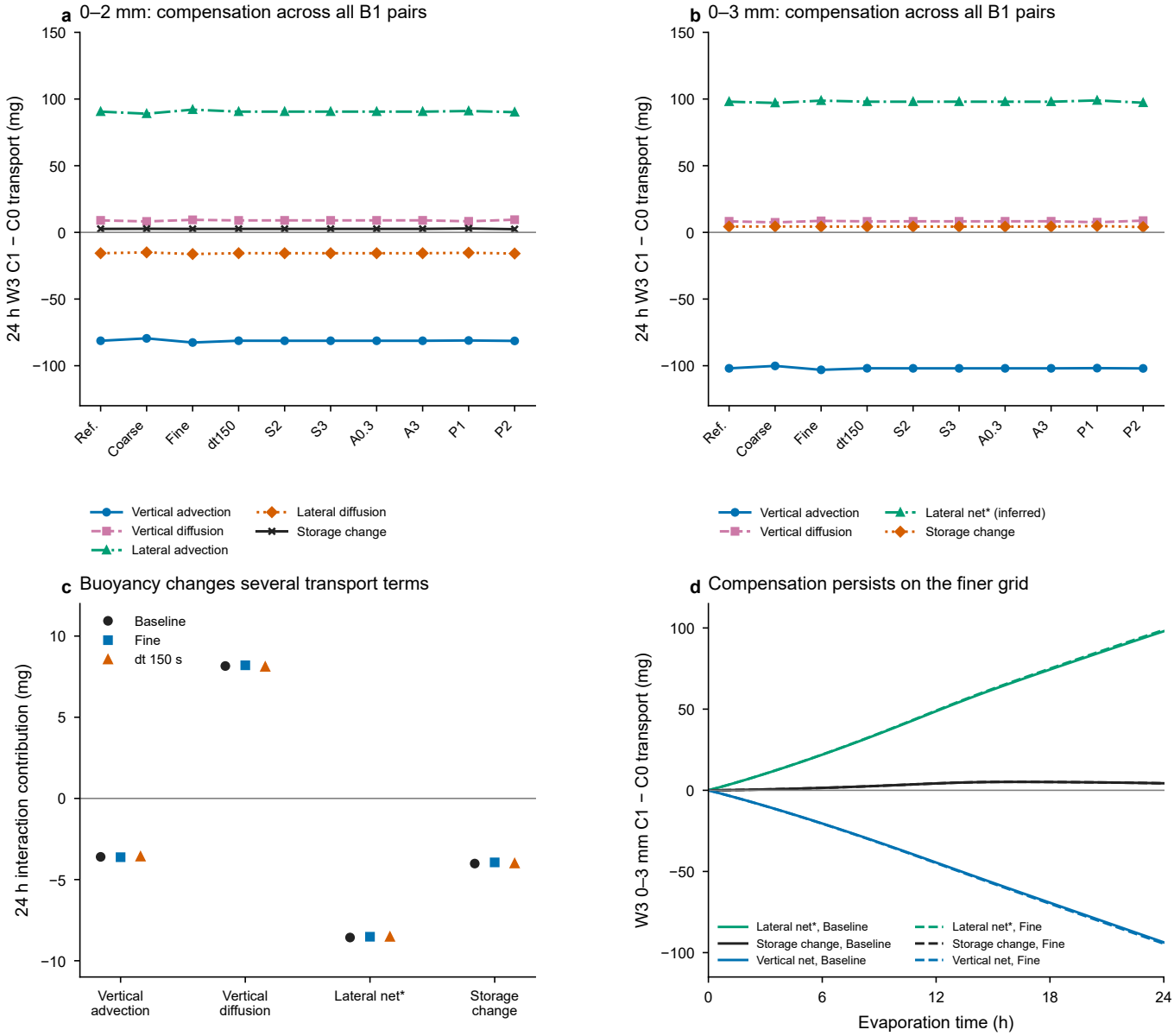


Upper 2 mm: independent lateral + vertical conservative flux balance. Upper 3 mm: *lateral NET is inferred from storage minus vertical input.
Vertical terms use face weights consistent with fractional-cell storage, rather than the nearest mesh plane. No lateral 0–3 mm mode split is fabricated.
Small storage differences can coexist with large opposing transport changes. This discrete-budget result does not establish all fluxes are grid independent.
S2/S3: phases 2/3; A0.3/A3: amplitude ×0.3/×3; P1/P2: conductivity corners. NaCl-equivalent mass at 58.443 g/mol; true 24 h endpoints.